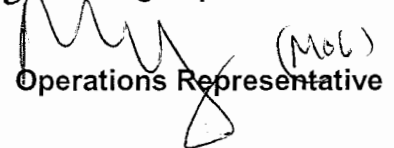


JOB PERFORMANCE MEASURE

STATION:	SALEM			
SYSTEM:	Administrative			
TASK:	Identify and Isolate Non-Essential Chilled Water Loads			
TASK NUMBER:	0980020202			
JPM NUMBER:	13-01 NRC SRO Admin A1-1			
ALTERNATE PATH:	☐	K/A NUMBER:	2.1.25	
		IMPORTANCE FACTOR:	4.2	
APPLICABILITY:		RO	SRO	
EO ☐	RO ☐	STA ☐	SRO ☒	
EVALUATION SETTING/METHOD:	Classroom			
REFERENCES:	S2.OP-SO.CH-0001 Rev. 30 (checked 9-25-14)			
TOOLS AND EQUIPMENT:	Calculator			
VALIDATED JPM COMPLETION TIME:	8 min			
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	N/A			
Developed By:	G Gauding Instructor	Date:	9-3-14	
Validated By:	N Salam SME or Instructor	Date:	10-16-14	
Approved By:	 Training Department	Date:	10-23-14	
Approved By:	 Operations Representative (MOC)	Date:	10-23-14	
ACTUAL JPM COMPLETION TIME:				
ACTUAL TIME CRITICAL COMPLETION TIME:				
PERFORMED BY:				
GRADE:	☐ SAT	☐ UNSAT		
REASON, IF UNSATISFACTORY:				
EVALUATOR'S SIGNATURE:			DATE:	

JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Chilled Water

TASK: Identify and Isolate Non-Essential Chilled Water Loads

TASK NUMBER: 0980020202

INITIAL CONDITIONS:

1. Salem Unit 2 is at 100% power during a summer heat wave.
2. At 1300, 22 Chiller trips on freeze protection followed shortly by 21 chiller, and neither can be reset.
3. Technical Specification LCO 3.7.10.a Action b.1. is in effect and non-essential heat loads need to be removed from the chilled water system.
4. Inlet Water Temperature Readings from SC.OP-DL.ZZ-0008(Q), Circulating / Service Water Log at 0800:
2TL3756 = 73.4°F
2TL3757 = 73.5°F
5. The past 2 days of Inlet Water Temperature Readings were:

	2TL2756	2TL3757
<u>Yesterday</u>		
0800	73.4	73.4
2000	74.2	74.2
<u>2 days ago</u>		
0800	73.3	73.3
2000	74.1	74.1
6. Outside air temperature has risen from 81°F to 99°F during this watch.

INITIATING CUE:

Identify non-essential heat loads to be removed from service in order to comply with TS 3.7.10. a Action b.1.

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made.

Task Standard for Successful Completion:

Identify combination of heat loads which will remove at least 999.1 kBTU/hr from chilled water system.

JOB PERFORMANCE MEASURE

System: Chilled Water

Task: Identify and Isolate Non-Essential Heat Loads

Name:

Date:

*	STEP NO.	STEP (* = Critical Step)	STANDARD	EVAL S / U	COMMENTS (Req'd for UNSAT Evaluation)
		Provide blank copies of TS 3.7.10, Chilled Water System – Auxiliary Building Subsystem and S2.OP-SO.CH-0001(Q), Chilled Water System Operation.			
	SO.CH-1 5.6.2	IF two Chillers are inoperable, THEN SELECT components for isolation IAW Attachment 2.			
* *	Att 2 1.0	RECORD the following data for use in Table A: 1.1 Inlet Water Temperature from SC.OP-DL.ZZ-0008(Q), Circulating / Service Water Log: 2TL3756= ____ °F 2TL3757= ____ °F From the above indicated 2TL3756/2TL3757 temperatures, RECORD the highest temperature AND ADD 1.5°F to account for instrument uncertainty. Utilize this temperature when referring to Table A ____ °F + 1.5°F = ____ °F (highest temp) 1.2 Number of inoperable Chiller Units: ____	Enters temperatures from cue sheet onto Attachment 2 form. Adds 1.5 deg. to highest temp (73.5) and enters 75.0 deg. Lists 2 chillers out of service.		

JOB PERFORMANCE MEASURE

System: Chilled Water

Task: Identify and Isolate Non-Essential Heat Loads

Name:

Date:

*	STEP NO.	STEP (* = Critical Step)	STANDARD	EVAL S / U	COMMENTS (Req'd for UNSAT Evaluation)
	Att 2 NOTE	Conservative decision making should be used when determining the appropriate non-essential heat loads to be isolated. If inlet water temperature is anticipated to exceed the selected inlet water temperature values, then heat loads should be removed for the next higher inlet water temperature values.	Based on the rise in outside air temperature between the 0800 inlet temperature reading and now (1300), and that the 75°F temp used for calculation is right at the limit to go up one category, determines heat loads for inlet temp >75°F should be used.		
*	2.0	RECORD the TOTAL HEAT LOAD ISOLATION from Table A: ____ kBTU/hr	Enters 999.1 kBTU/hr.		
	3.0	IDENTIFY the components to be isolated in Table B as follows: 3.1 RECORD the value from the HEAT LOAD column into the Isolation column for the components selected for isolation.	Records values from HEAT LOAD into Isolation column in Table B for following components: Any combination of components may be selected as long as the total heat load selected is > 999.1 kBTU/hr.		
*	4.0	ADD the values recorded in the Isolation column AND RECORD the Total Isolation value in Table B	Adds values in ISOLATION column. Verifies total is greater than 999.1 kBTU/hr.		

JOB PERFORMANCE MEASURE

System: Chilled Water

Task: Identify and Isolate Non-Essential Heat Loads

Name:

Date:

*	STEP NO.	STEP (* = Critical Step)	STANDARD	EVAL S / U	COMMENTS (Req'd for UNSAT Evaluation)
	5.0	ENSURE that the Total Isolation value recorded in Table B is $\geq$ the Total Heat Load Isolation recorded in Step 2.0 of this attachment.	Verifies Total Isolation value recorded in Table B is $\geq$ the Total Heat Load Isolation recorded in Step 2.0 of this attachment. Terminate JPM when procedure SO.CH-0001(Q), Chilled Water System Operation is returned to evaluator.		

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11/18/14

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JOB PERFORMANCE MEASURE

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- VM 1. Task description and number, JPM description and number are identified.
- VM 2. Knowledge and Abilities (K/A) references are included.
- VM 3. Performance location specified. (in-plant, control room, or simulator)
- VM 4. Initial setup conditions are identified.
- VM 5. Initiating and terminating Cues are properly identified.
- VM 6. Task standards identified and verified by SME review.
- VM 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- VM 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 30 Date 5/19/2011
- VM 9. Pilot test the JPM:
a. verify Cues both verbal and visual are free of conflict, and
b. ensure performance time is accurate. @ 8 min
- N/A 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- N/A 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: MASSA Salam

Date: 10/16/14

SME/Instructor: _____

Date: _____

SME/Instructor: _____

Date: _____

JOB PERFORMANCE MEASURE

INITIAL CONDITIONS:

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